🌿 **Project Title: MedShare – Medicine Expiry & Donation System**

🎯 **Objective**

To minimize medicine wastage and help underprivileged communities by connecting donors, NGOs, and hospitals through a digital donation and tracking system.

🧰 **Technology Stack**

Layer	Tools
Frontend	HTML5, CSS3, JavaScript (ES6), Bootstrap / TailwindCSS
Backend	Python, Django Framework
Database	SQLite / MySQL
APIs	Google Maps API (for location-based features), Email/SMS APIs
Others	Django Admin, Django ORM, Authentication System, REST Framework (optional)

🗂️ **Website Structure (Pages & Modules)**

🔹 **1. Home Page (/)**

Purpose: First impression of the platform.
Features:

- Navbar with links (Home, About, Donate, NGOs, Contact, Login/Signup)
- Hero section: slogan like "Don't Waste, Donate – Save Lives with MedShare"
- "How It Works" section (Step-by-step: Register → List Medicines → NGO Pickup)
- Real-time counter (Medicines Donated, NGOs Registered, People Helped)
- Testimonials & success stories section
- Footer with contact info, social links

🔹 **2. User Authentication Pages**

Pages:

- Signup Page (Choose Donor / NGO / Admin)
- Login Page
- Forgot Password

Features:

- Role-based registration (User type: Donor / NGO / Hospital)
- Email verification after registration
- Django authentication system with secure password handling

🔹 3. Donor Dashboard (/donor/dashboard)

Features:

- View Profile (Name, Address, Email, Contact)
- Add New Medicine Donation Form:
  - Medicine name, quantity, expiry date, image, location
- View List of Donated Medicines
- Edit/Delete donations
- Expiry alerts: "Medicine XYZ is expiring in 10 days"
- Track donation status (Pending / Accepted / Collected)
- Notification system (email or in-app)

🔹 4. NGO / Hospital Dashboard (/ngo/dashboard)

Features:

- View list of available medicines (filtered by city/location)
- Search / filter by medicine name, expiry date
- Request medicine pickup from donor
- Accept or reject donations
- View donation history
- Live Map View: Nearby donors

🔹 5. Admin Panel (/admin/)

Features:

- Manage users (Donors / NGOs)
- Approve new NGO registrations
- Monitor donation activities
- Generate reports (e.g., number of medicines donated, most active donors)
- Manage notifications & alerts

🔹 6. About Page (/about)

Features:

- Purpose and mission of MedShare
- Statistics about medicine wastage in India/worldwide
- Team and contributors section
- Link to government or NGO partners

🔹 7. Contact Page (/contact)

Features:

- Contact form (Name, Email, Message)
- Google Map showing MedShare HQ or nearest NGO
- Auto-email reply feature using Django Email backend

🔹 8. FAQ / Help Page (/faq)

Features:

- Common questions like:
  - "What medicines can I donate?"
  - "How does MedShare ensure safety?"
- Step-by-step donation guide

🔹 9. Real-Time Expiry Tracker (/tracker)

Features:

- Displays all user medicines with expiry countdown
- JS-based timer for each item (e.g., "5 days left before expiry")
- Automatic email reminder before expiry

🔹 10. Reports & Analytics (/reports)

For Admin only
Features:

- Graphs and charts (via Chart.js or Recharts)
  - Donations per month
  - NGO participation rate
  - Medicines saved from wastage
- Export report as PDF or CSV

💡 Unique & Innovative Features

✅ AI-based expiry alert using Python date difference logic
✅ Google Maps integration for real-time donor–NGO matching
✅ Secure user authentication and data management
✅ Email notifications for donation confirmation and reminders
✅ Real-time analytics dashboard for Admin
✅ Sustainable and social impact-driven solution

🌍 Social Impact

- Reduces wastage of valuable medicines
- Provides timely help to the poor and underprivileged
- Encourages responsible medicine management
- Strengthens collaboration between citizens and NGOs

🧱 Suggested Database Tables

Table	Key Columns
User	user_id, name, email, password, role (donor/ngo/admin), address, contact
Medicine	med_id, user_id (FK), name, quantity, expiry_date, image, status
DonationRequest	request_id, med_id (FK), ngo_id (FK), status (pending/accepted/delivered)
Notification	notif_id, user_id (FK), message, created_at
ContactForm	id, name, email, message

🎨 Frontend Suggestions

- Use **Tailwind CSS** or **Bootstrap 5** for clean design
- Add **Font Awesome** icons for modern UI
- Use light green & white theme (symbol of health and care)
- Add animations using **AOS (Animate On Scroll)** or simple JS effects

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🚀 Future Enhancements

- Add mobile app using React Native or Flutter
- Integrate AI model to detect expired medicines from images
- Connect with verified government health databases
- Implement donation certificate generator for donors